

Executive Summary

CulturaLens

A Human-Centered Evaluation Framework for Cultural Localization in AI Image Generation for India

Problem

Modern image generation models produce highly realistic images, but **realism does not guarantee cultural accuracy**.

Current evaluation benchmarks primarily measure image quality and prompt adherence, leaving a critical gap in understanding **how well AI represents India's regional cultures**.

This becomes especially important for education, where culturally inaccurate visuals can reduce learner trust and reinforce incorrect representations.

Objective

Develop a **human-centered evaluation framework** to benchmark how effectively frontier AI image models understand and represent India's cultural diversity.

The evaluation focused on whether models could accurately generate:

- State-specific festivals
 - Traditional attire
 - Regional cuisine
 - Household environments
 - Architecture
 - Cultural symbols
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Why This Evaluation Matters

Why Education?

- Students are among the most trust-sensitive users.
- Visual inaccuracies directly impact learning quality.
- AI-generated educational content is expected to become increasingly common.

Why India?

- India represents one of the world’s most culturally diverse environments.
- Strong performance on generic benchmarks does not necessarily translate to successful localization.

Why AI Labs Should Care

- Cultural understanding is becoming a key product differentiator.
- Localization quality directly impacts deployment readiness across education, marketing, and public communication.

Evaluation Design

Component	Value
Models Evaluated	GPT Image 1, Gemini 3.1 Flash Image Preview, Gemini 2.5 Flash Image
States Covered	10
Images Generated	30
Participants	8
Evaluation Type	Blind Human Evaluation + Objective Benchmark

Prompt Strategy	Identical Zero-Shot Prompt
Generation Policy	First Generation Retained

Evaluation Framework

Every image was evaluated using three complementary layers.

Human Evaluation

- State Representation
- Photorealism
- Prompt Compliance
- Educational Suitability

Objective Benchmark

- Prompt Compliance
- Cultural Consistency
- Technical Quality

Failure Analysis

- Generic India Bias
 - Cultural Leakage
 - Localization Precision
 - Cultural Density
 - Common Sense Reasoning
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Key Findings

GPT Image 1 delivered the strongest overall performance.

- Highest participant preference.
- Best photorealism.

- Strongest prompt adherence.
 - Most suitable for educational deployment.
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Gemini 3.1 demonstrated a substantial improvement over Gemini 2.5.

- Better regional representation.
 - Improved visual quality.
 - Stronger cultural consistency.
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Photorealism has largely converged.

The primary challenge has shifted from generating realistic images to generating **culturally accurate** images.

Uttar Pradesh emerged as the most difficult benchmark.

Across all three models:

- Highly generic outputs.
 - Weak regional identity.
 - Frequent confusion with other North Indian states.
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Five recurring localization failures were consistently observed.

- Generic North Indian representation
 - Cultural leakage across states
 - Incorrect festival and religious context
 - Repetitive decorative stereotypes
 - Common-sense inconsistencies
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Product Outcome

Developed **CulturaLens**, a scalable evaluation framework consisting of:

- Structured evaluation rubric

- Blind human benchmark
- Objective scoring methodology
- Failure mode taxonomy
- Interactive evaluation dashboard
- State-wise comparison framework

The framework is designed to help AI labs systematically benchmark localization quality, identify recurring failure patterns, and prioritize future model improvements.

Strategic Recommendation

Current AI image benchmarks should evolve beyond measuring **visual realism** and begin systematically evaluating **cultural localization**.

For culturally diverse markets such as India, localization quality will increasingly determine user trust and product adoption.

Final Takeaway

Photorealism is no longer the bottleneck. Cultural localization is.

The next generation of AI systems should be evaluated not only by how realistic they appear, but by **how faithfully they represent the people and cultures they are designed to serve.**